MIL-A-44322(GL)
19 February 1988

MILITARY SPECIFICATION

APPLE PIE, SEMI-SOLID, THERMOSTABILIZED TUBE PACK

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers tube packed semi-solid apple pie for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44320 - Packaging and Processing of Foods in Collapsible Containers (Tubes)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Grade A Pasteurized Milk Ordinance

(Application for copies should be addressed to the Department of Health and Human Services, U.S. Public Health Service, Food and Drug Administration, Milk Safety Branch, Code HFF-346, 200 C Street S.W., Washington, DC 20204.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Grades of Canned Applesauce

(Application for copies, should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Applesauce, canned, sweetened. Canned applesauce shall be sweetened, natural color and flavor type and regular (comminuted) style. The canned applesauce shall be U.S. Grade A of the U.S. Standards for Grades of Canned Applesauce and shall be of the latest season's pack.

3.2.2 Milk. Milk shall comply with 21 CFR Part 131.110 and shall be Grade A pasteurized in accordance with the applicable requirements of the Grade A Pasteurized Milk Ordinance. The milk shall be homogenized and shall have a clean fresh flavor, shall be free from objectionable flavors and odors, and shall be not older than 72 hours at time of product preparation (see 3.3.2).

3.2.3 Sugar, brown, light. Brown sugar shall be partially refined cane or beet sugar. The sugar shall be light brown in color and shall possess a sweet, molasses-like flavor.

3.2.4 Cream, heavy. Heavy cream shall comply with 21 CFR Part 131.150 and shall be pasteurized in accordance with the applicable requirements of the Grade A Pasteurized Milk Ordinance. The heavy cream shall be homogenized and shall have a clean fresh flavor, shall be free from objectionable flavors and odors, and shall be not older than 72 hours at time of product preparation (see 3.3.2).

3.2.5 Juice, lemon. Lemon juice shall be canned or frozen, single strength or concentrated. The lemon juice shall be prepared from unfermented juice obtained from fresh, sound, mature lemons of one or more of the high acid varieties.

3.2.6 Sugar, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet, or a combination of both and shall comply with MIL-STD-900.

3.2.7 Eggs, whole, liquid or frozen. Whole eggs shall be liquid or frozen and shall have been processed under continuous USDA inspection in accordance with USDA Regulations Governing the Inspection of Eggs and Egg Products. Whole eggs shall be prepared from egg yolks and egg whites in their natural proportions as broken directly from the shell egg. The whole eggs shall contain no chemical processing additives and shall be USDA Grade A. Egg products shipped between plants shall be accompanied by a USDA Egg Products

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Inspection Certificate. Liquid egg products shall be held at a temperature of 40°F or lower and shall be held not more than 72 hours from time of pasteurization until the start of formulation. Frozen egg products shall be held at a temperature of 10°F or lower and used within 120 days.

3.2.8 Starch, waxy, maize, modified. Starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and shall have a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.9 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.10 Margarine. The margarine shall conform to the FDA Standard of Identity for Margarine (21 CFR Part 166.110), shall possess a sweet, pleasing taste and aroma, shall have a medium yellow color, and shall have a smooth body and a texture free from crumbliness, graininess, grittiness, oiliness, or droplets of free moisture.

3.2.11 Cinnamon, ground. Ground cinnamon shall be prepared from the dried bark of Cinnamomum burmanli Blume, commonly known as "Korintz Cinnamon." The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.12 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride as described in the Food Chemicals Codex.

3.2.13 Nutmeg, ground. The ground nutmeg shall be derived from the dried kernel of Myristica fragrans Houtt. The nutmeg shall have a dull brown color. The nutmeg shall contain not less than 7.5 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.14 Flavoring, vanilla. Vanilla flavoring shall be pure vanilla extract.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formulation.

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3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Product formulation. The formula for semi-solid apple pie for tube pack shall be as follows:

Ingredients	Percent by Weight
Applesauce	36.00
Milk, whole, pasteurized	20.65
Sugar, light brown	10.60
Cream, heavy	10.40
Juice, lemon	5.10
Sugar, granulated	4.80
Eggs, whole	3.40
Starch, waxy, maize, modified	3.30
Water for starch slurry 1/	3.30
Margarine	1.60
Cinnamon, ground	0.40
Salt	0.40
Nutmeg	0.04
Vanilla flavoring	0.01

1/ The total amount of water used in the starch slurry shall be adjusted as necessary to adjust the product viscosity and facilitate pumping and filling.

3.3.2 Product preparation. All ingredients except starch, water, and eggs shall be combined in a steam jacketed kettle and heated to 165° to 180°F. The mixture shall be constantly stirred until all the margarine is completely melted. A slurry shall be prepared by mixing the water and starch. The starch slurry shall be added to the mixture and the mixture shall be reheated to 165° to 180°F and held in this temperature range for approximately 10 minutes. A small amount of the hot mixture shall be slowly added to the eggs to help prevent coagulation. The egg mixture shall be added to the ingredient mixture and mixed well to form the product. The product shall be maintained in the temperature range of 150° to 180°F and filled into the tubes within 2 hours after preparation.

3.4 Tube filling and sealing. Five ounces of product shall be filled into a tube as specified in MIL-P-44320. The filled tubes shall be immediately given a saddle-back fold and crimped securely, then sealed under vacuum by flushing the head space of each filled tube with super-heated steam. The filled and sealed tubes shall be thermoprocessed to an F_o value of 6 within 2 hours after sealing in accordance with MIL-P-44320.

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3.5 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, moldy, or sulfurous.

c. The product shall be brownish in color.

d. The product shall have a characteristic apple pie flavor.

e. The average net weight shall be not less than 5.0 ounces.

f. No individual tube shall contain less than 4.7 ounces of product.

g. The salt content shall be not greater than 0.75 percent nor less than 0.65 percent.

h. The product shall not be rubbery.

i. The product shall be smooth, without lumps, and shall not be excessively thick or thin.

j. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.5.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample in palatability and overall appearance (see 4.4, 6.1, and 6.2).

3.6 Plant qualifications. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

3.7 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, and filling requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed tube inspection. Inspection of filled and sealed tubes shall be in accordance with the quality assurance provisions of MIL-P-44320.

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4.5.4 Net weight inspection. The net weight of the filled and sealed tubes shall be determined by weighing each sample unit on a suitable scale tared with a representative empty tube. Any individual net weight of less than 4.7 ounces shall be scored as a minor defect. The lot size shall be expressed in tubes. The sample unit shall be one filled and sealed tube. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 5.0 ounces.

4.5.5 Product inspection. The filled and sealed sample tubes shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the retort process. The tubes shall be opened and inspected for the defects indicated in table I. The lot size shall be expressed in tubes. The sample unit shall be the contents of one tube. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE I. Product defects 1/ 2/

Category	Defect
Major	
101	Product is rubbery
102	Product is not smooth
103	Product is excessively thick or thin
104	Product is lumpy
105	Product shows evidence of excessive heating (materially darkened or scorched)
106	Product does not have a characteristic apple pie flavor
107	Product does not have a characteristic brownish color

1/ The presence of foreign material (e.g., dirt, insects, insect parts, hair, wood, paper, paint, glass, metal), foreign odor or flavor (e.g., burnt, scorched, stale, sour, rancid, musty, moldy, sulfurous), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparision shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

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4.5.6 Salt content testing. The sample tubes shall be opened and the entire contents of each tube shall be separately blended in a Waring blender or equivalent. The salt content of the product from each tube in the sample shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. Test results shall be reported to the nearest 0.01 percent. Any result failing to conform to the salt requirement in 3.5 shall be classified as a major defect. The lot size shall be expressed in tubes. The sample unit shall be one filled and sealed tube. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Packaging inspection shall be in accordance with the quality assurance provisions of MIL-P-44320.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the applicable requirements of MIL-P-44320.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.5.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Apple
Pie
Semi-Solid
Tube

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Custodian:

Army - GL

Review activities:

Army - MD, TS
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

Project No. 8940-A591

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-44322(GL)	Apple Pie, Semi-Solid, Thermostabilized Tube Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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